

Identifying NHL Player Archetypes Using PCA and K-Means Clustering on NHL EDGE Tracking Data

Problem Definition

Roster construction and shift-by-shift deployment in professional hockey are still driven mostly by intuition and by raw counting stats. Coaches and general managers talk about "the eye test," and most public-facing player evaluations rely on goal and point totals that don't really describe how a player generates those numbers or how they spend the majority of their ice time. We set out to do two things. First, use unsupervised learning on three seasons of NHL skater data, including the league's new EDGE tracking feed, to recover a small set of player archetypes that line up with the kinds of roles a coach would actually recognize. Second, test whether those archetypes differ in measurable on-ice impact at even strength, on the power play, and on the penalty kill. Critically, the metrics we use to evaluate the clusters (goal rates, goal share, shot-attempt share) are held out of the clustering itself, so any difference we find between archetypes reflects the playstyle structure the algorithm picked up rather than the outcomes we hoped to see.

Description of Background

Hockey is a continuous, low-event sport, and box-score totals tend to flatten the context that produced them. A player who racks up hits might be a punishing defensive forward, or might just be on the ice while their team chases the puck. Sports analytics has spent the last two decades pushing back on this kind of context-free counting, and clustering work in basketball and soccer has shown that unsupervised methods can recover the role groupings that scouts already use, in a reproducible way. Hockey has lagged that trend in part because puck-tracking data only became available league-wide with the rollout of NHL EDGE, which captures skating speed, shot speed, and shot location at a granularity traditional play-by-play data could not. Our contribution is to apply PCA and K-Means to a feature set that combines standard rate stats with these tracking metrics, and then to test the resulting clusters against game-state-specific outcomes — not just describe the cluster centroids the way most public clustering writeups do. That second step is what lets the analysis speak to the deployment question rather than only to the labeling question.

Description of Dataset

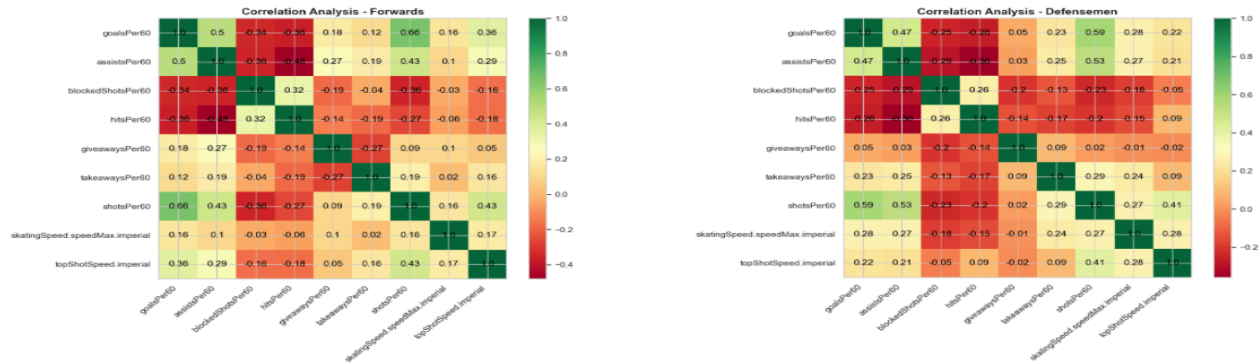
We pulled three regular seasons (2023–24 through 2025–26) from the public NHL API using the `nhl-api-py` module. The acquisition pipeline scraped six different report endpoints — summary, biography, time-on-ice, shooting, percentages, and real-time tracking — using paginated queries, and then a separate multithreaded job (ten concurrent workers) hit the EDGE endpoint once per player-season for skating speed, shot speed, and shot location. The EDGE step required over 2,700 individual requests and was the slowest part of the pipeline. Nested shot-location records were parsed into engineered features for high-, medium-, and long-danger shot rates. The raw join produced about 2,754 player-season rows across more than 100 columns.

Cleaning was more involved than expected. A subtle issue with the EDGE feed is that when a tracking camera fails or a measurement is unavailable, the API returns 0.0 instead of null. If left in, this would have made K-Means cluster players by which arenas had broken cameras rather than by playstyle. We replaced these pseudo-zeros with NaN on the affected tracking columns and then dropped any row missing a clustering input. We also filtered out players with fewer than ten games played, since rate stats over a few games are dominated by noise. Counting stats were converted to per-60-minute rates for overall and even-strength categories, and to per-2-minute rates for power play and penalty kill (the latter approximating one special-teams shift). After these steps the working dataset held 2,311 player-season rows and 162 columns. We modeled forwards and defensemen separately because their role distributions are fundamentally different — defensemen average about 4.6 blocks per 60 against 2.2 for forwards, and forwards average roughly 0.77 goals per 60 against 0.21 for defensemen — and clustering them together would force the model to spend most of its variance on the position label we already know. After dropping rows missing any tracking input, the final clustering inputs were 997 forward-seasons and 532 defenseman-seasons.

Description of Methods Used

Feature selection and correlation. We started from more than 100 candidate columns and narrowed to seventeen features covering scoring (`goalsPer60`, `assistsPer60`, `shotsPer60`, `shootingPct5v5`), physicality (`hitsPer60`, `penaltyMinutesPer60`), defensive contribution (`blockedShotsPer60`, `takeawaysPer60`, `giveawaysPer60`), shot quality and location (`highDangerShotsPer60`, `longDangerShotsPer60`), athletic profile from the EDGE feed (`skatingSpeed.burstsOver20`, `topShotSpeed.imperial`), and deployment context (`zoneStartPct5v5`, `ppTimeOnIcePctPerGame`, `shTimeOnIcePctPerGame`). Forwards additionally received `faceoffWinPct`, which is not meaningful for defensemen. Pairwise correlation showed the structure

we expected: the offensive features cluster tightly together (goals and shots correlate at $r \approx 0.66$ for forwards), the offensive and defensive features correlate negatively ($r \approx -0.36$), and the EDGE tracking metrics correlate weakly with everything else ($r < 0.20$), which means they contribute genuinely independent information rather than restating the box score.

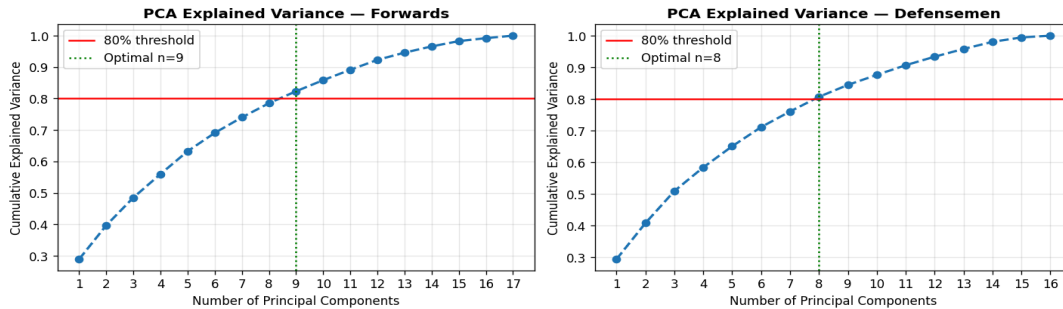


Figures 1, 2: Correlation Heat Maps for Forwards and Defensemen respectively

PCA. K-Means uses Euclidean distance, so correlated input features double-count whatever underlying trait they share. We standardized every feature with StandardScaler and then ran PCA before clustering, retaining the smallest number of components that explained at least 80% of total variance: 9 components for forwards (82.3% cumulative) and 8 for defensemen (80.7%). The resulting components turned out to be readable. PC1 in both models is essentially an offensive-deployment axis — it loads heavily positive on goals, shots, and PP TOI, and heavily negative on hits and blocks — and later components separated penalty-kill usage, takeaway-vs-giveaway puck management, and shot location preference. This is the largest methodological change from our progress report, where we used a smaller raw feature set directly. PCA gave us cleaner silhouette scores and, more importantly, made the cluster axes interpretable as orthogonal playstyle dimensions instead of collinear stat bundles.

K-Means. We ran K-Means on the principal components for k from 3 to 8 with n_init=10, scored each value with the silhouette coefficient, and picked the k that maximized that score. Both forwards and defensemen reached their maximum at k=4 (forward silhouette 0.128, defenseman 0.150). Those values are modest in absolute terms, but that's expected for unsupervised clustering of human behavior — players exist on a continuum, not in fully separated groups — and the four-cluster solution was clearly preferred over both coarser and finer splits. Cluster labels were

assigned by inspecting the unscaled feature averages within each cluster (Tables 1 and 2) and matching them to recognizable hockey roles.



Figures 1, 2: Cumulative PCA variance for forwards (9 components, 82%) and defensemen (8 components, 81%).

Experiment: Setup and Results

Tables 1 and 2 give the unscaled mean values that anchor each archetype. Figures 3–8 then evaluate whether those styles actually drive different on-ice outcomes.

Table 1. Forward archetype profiles (mean unscaled values).

Stat	Elite Off. Engine	Two-Way / Sec. Scorer	Physical Enforcer	Defensive Specialist
N (player-seasons)	215	374	115	293
goalsPer60	1.26	0.81	0.50	0.46
assistsPer60	1.64	1.07	0.57	0.69
shotsPer60	8.97	6.96	5.87	5.08
hitsPer60	3.02	4.10	16.09	6.74
blockedShotsPer60	1.67	1.95	2.60	2.88
pp TOI share	0.61	0.35	0.04	0.08
sh TOI share	0.12	0.10	0.05	0.28
zoneStart % (off.)	0.56	0.52	0.44	0.40
speed bursts >20	139.7	80.9	49.9	80.0

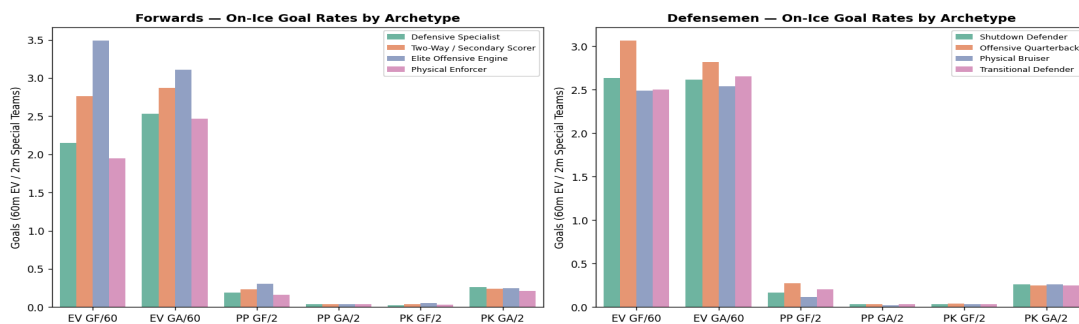
Forward archetypes. **Elite Offensive Engines** (n=215) lead in every offensive category, take a 61% share of available power-play time, and post the highest speed-burst counts (~140). The McDavid-, Draisaitl-, Ovechkin-tier top-line skaters. **Two-Way / Secondary Scorers** (n=374) are the largest group and produce moderate offense (0.81 G/60, 1.07 A/60) with balanced physicality: middle-six forwards who carry most of a team's depth scoring. **Physical Enforcers** (n=115) are defined almost entirely by hits (16.1 per 60, ~4× the league average) and PIM (5.9 per 60), with the lowest skating-burst counts of any forward group. **Defensive Specialists** (n=293) are the only forward archetype with substantial penalty-kill deployment (28% sh TOI share), modest offense, and the lowest offensive zone start rate (0.40).

Table 2. Defenseman archetype profiles (mean unscaled values).

Stat	Off. Quarterback	Transitional Defender	Shutdown Defender	Physical Bruiser
N (player-seasons)	116	158	208	50
goalsPer60	0.38	0.19	0.13	0.16
assistsPer60	1.21	0.70	0.60	0.53
shotsPer60	5.77	3.77	3.15	3.84
hitsPer60	2.41	3.03	4.43	9.34
blockedShotsPer60	4.01	4.17	5.49	4.94
pp TOI share	0.47	0.16	0.04	0.03
sh TOI share	0.29	0.16	0.42	0.26
zoneStart % (off.)	0.55	0.54	0.44	0.47
speed bursts >20	65.2	26.7	27.6	25.0

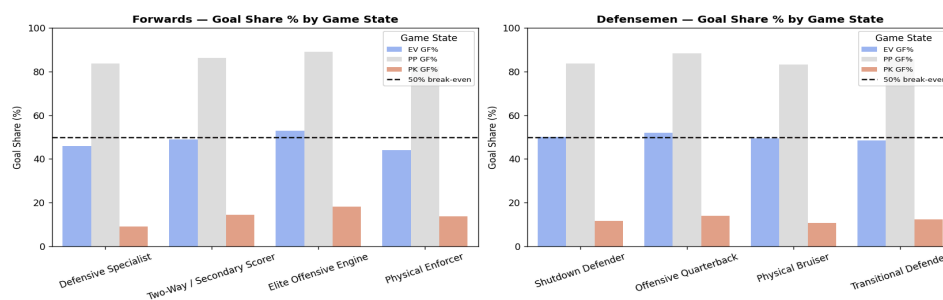
Defenseman archetypes. **Offensive Quarterbacks** (n=116) are the elite puck-movers. 1.21 A/60, 5.77 SH/60, a 47% PP TOI share, and the only defenseman group with high speed-burst counts (65 vs ~26 for the others). Cale Makar, Quinn Hughes, and Adam Fox all fall here. **Shutdown Defenders** (n=208) are the largest group, defined by the heaviest penalty-kill deployment (42% sh TOI share) and the highest blocked-shot rate (5.49). **Physical Bruisers** (n=50) are a small group whose hit rate (9.34) is more than double any other defenseman cluster and whose PIM is over 3× the others'. **Transitional Defenders** (n=158) are the cluster PCA exposed most clearly. They post shutdown-level blocked shots (4.17) but markedly higher assist rates (0.70) than the pure shutdown group, with modest PP usage (16% TOI share). They look like a hybrid two-way role that hockey commentary often doesn't name as a distinct type, and we did not see them split out cleanly in our progress report, before the PCA step.

We then evaluated the four-cluster solution against three independent outcome metrics that were not used during clustering: on-ice goal rates by game state, goal share percentages, and average shot-attempt share (SAT%). Holding outcomes out of the input set lets us check whether the clusters reflect playstyles that actually move results, rather than stylistic differences with no on-ice impact.



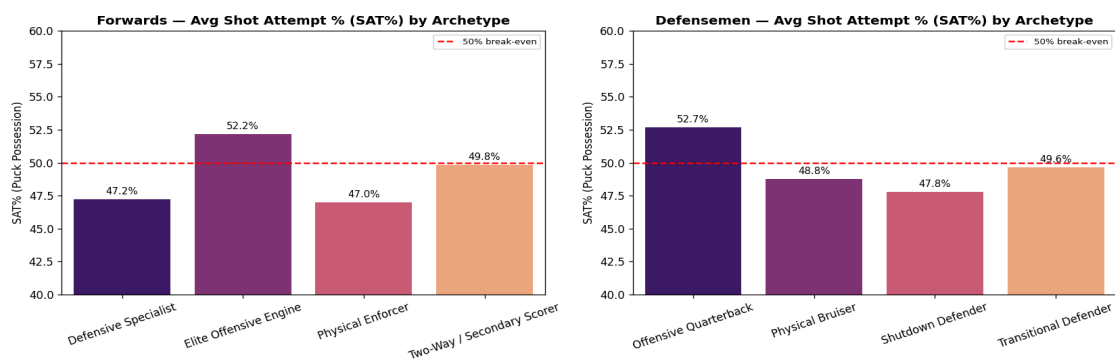
Figures 3, 4: On-ice goal rates by archetype. EV stats per 60 minutes; PP/PK stats per 2 minutes (~one special-teams shift).

On the forward side, Elite Offensive Engines drive the strongest results in every direction at once: 3.49 EV GF/60 (vs 2.15 for Defensive Specialists), 0.30 PP GF/2 (highest of any group), and a 52.9% even-strength goal share — the only forward archetype above 50% at 5-on-5. **This is the clearest single finding of the project.** Whatever else a team does, the on-ice numbers say that having an elite scorer on the ice is the largest controllable lever for goal differential. The Two-Way / Secondary Scorer group sits very close to break-even (49.1% EV GF%), Defensive Specialists are well below it (45.9%), and Physical Enforcers post the lowest share of any group (44.2%) — they also concede nearly as many even-strength goals (2.46 GA/60) as they score (1.95 GF/60). The penalty-kill numbers tell the story from the other side: Defensive Specialists allow 0.21 PK goals against per 2 minutes, the lowest of any forward group, exactly as expected from a cluster that absorbs 28% of available short-handed time.



Figures 5, 6: Goal share by archetype at each game state. The 50% line marks break-even at even strength.

On the defenseman side the rates are flatter at even strength but the special-teams differences are pronounced. Offensive Quarterbacks are the only defenseman group above 50% EV GF share (52.1%) and post the highest PP GF/2 (0.27 vs 0.11 for Physical Bruisers). The Transitional Defender result is what we found most interesting. **These players don't look like classic puck-movers** — their PP TOI share (0.16) is well below the Quarterback group's (0.47) — but they post a 49.7% SAT% and an 85.8% PP goal share when they are out there, both noticeably better than the pure shutdown group, while still carrying respectable blocked-shot numbers. They appear to be playing a quietly effective two-way role that doesn't show up in either the offensive-leader or defensive-leader columns of a standard scoring summary, and the PCA step is what made them visible. Shutdown Defenders are the most heavily used penalty killers by a wide margin (42% sh TOI share) but allow PK goals at essentially the same rate (0.26 per 2 min) as the other defenseman groups, which suggests their PK value is more about being trusted with the volume of usage than about a measurable per-shift edge.



Figures 7, 8: Average shot-attempt share (SAT%) by archetype. SAT% removes most shooting and goaltending variance from the goal-share picture.

Figures 7 and 8 confirm the goal-share findings using a metric much less sensitive to short-run shooting and goaltending variance. Both Elite Offensive Engines (52.2%) and Offensive Quarterbacks (52.7%) clear the 50% line; Physical Enforcers (47.0%) and Shutdown Defenders (47.8%) sit below it. The agreement between the SAT% and goal-share rankings means the playstyle differences we are seeing aren't an artifact of one season's shooting luck.

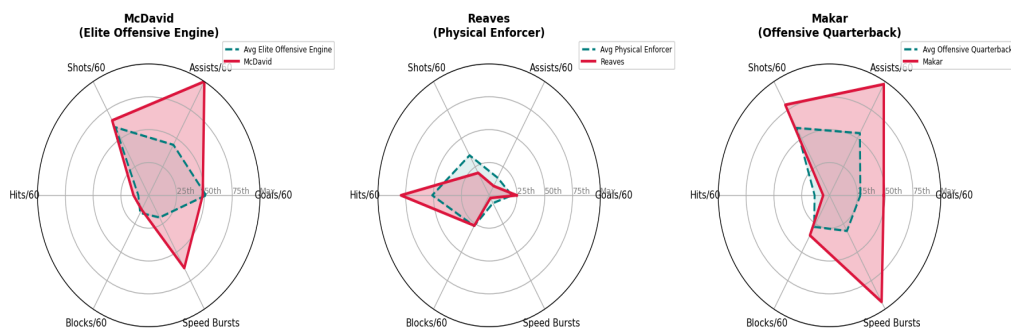


Figure 9: Player DNA radar charts. Each plot overlays an individual player (crimson) on the average normalized profile of their assigned archetype (teal). Stats are MinMax-scaled within position.

Figure 9 shows how the same archetype labels can be applied at the individual scouting level. Connor McDavid sits at or near the maximum of his cluster on goals, assists, and shots — he's an extreme example of his Elite Offensive Engine archetype rather than a borderline case. Ryan Reaves is a clean fit for Physical Enforcer: hits dominate his profile and almost everything else is well below his cluster average. Cale Makar is well above his Offensive Quarterback baseline on goals, assists, and speed bursts — an elite-forward-level offensive workload from the blue line. The general lesson is that the K-Means archetype gives the baseline expectation for a role, and the most valuable players in any cluster are the ones whose individual profile pushes well past that baseline on the dimensions that matter for the role.

Observation and Conclusion

Three findings stand out. First, the K-Means archetypes have measurable on-ice consequences. Elite Offensive Engines (forwards) and Offensive Quarterbacks (defensemen) are the only groups above 50% EV goal share and above 50% SAT%, and Elite Offensive Engines are the largest separator in even-strength goal differential: close to a full goal per 60 minutes higher than any other forward group. The proposal hypothesis (offensive types should dominate the power play, defensive types should do better on the penalty kill) is supported, but the more practically useful observation is that elite scorers help on every game state at once, and that finding is robust to switching from goals to a possession-based metric like SAT%.

Second, running PCA before K-Means changed the defenseman solution in a way our progress report did not capture: the algorithm cleanly separated a Transitional Defender cluster previously absorbed into the broader shutdown group, because multicollinearity in the offensive features had pulled those players toward whichever neighboring cluster was closest in raw Euclidean space. Preprocessing matters for clustering interpretability, not just numerics. Third, Figure 9 suggests these archetypes are most useful as a baseline for a role rather than a ceiling: the cluster average benchmarks what a typical player delivers, and deviation on the dimensions that matter is what separates elite players from replacement-level ones. Limitations include modest absolute silhouette scores (which reflect the continuum nature of playstyle), uneven NHL EDGE coverage in older seasons that forced us to drop several hundred player-seasons, and residual season-to-season noise in goal rates. Future work could replace K-Means with Gaussian mixtures to give per-player archetype probabilities (better suited to hybrid players like the Transitional Defenders), incorporate expected-goals models, and parse shift-by-shift play-by-play data to study direct matchups between archetypes.

References

- [1] Zmalski. "NHL-API-Reference." GitHub, 2024. <https://github.com/Zmalski/NHL-API-Reference>
- [2] A. Chamberlain. "Hot Hand or Hard Data: Is Recent Performance Weighted Enough in Pitching Projections?" FanGraphs, 2023. <https://fantasy.fangraphs.com/hot-hand-or-hard-data-is-recent-performance-weighted-enough-in-pitching-projections/>
- [3] V. Singh. "Multicollinearity in Regression: A Guide for Data Scientists." DataCamp, 28 Oct. 2024. <https://www.datacamp.com/tutorial/multicollinearity>
- [4] FlyRank. "Leveraging K-Means Clustering for Sports Analytics." 25 Apr. 2025. <https://flyrank.zendesk.com/hc/en-us/articles/26277513527698-Leveraging-K-Means-Clustering-for-Sports-Analytics>
- [5] I. T. Jolliffe and J. Cadima. "Principal component analysis: a review and recent developments." Phil. Trans. R. Soc. A, vol. 374, no. 2065, 2016.